Suppression of the quadrupole-misalignment false EDM signal by orbit-based methods in a proton EDM storage ring, and its limits

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In the proposed symmetric-hybrid storage ring for measuring the proton electric dipole moment (EDM), micron-scale transverse misalignments of the focusing quadrupoles generate a vertical spin precession that mimics a genuine EDM. This "false EDM" arises as a geometric (Berry) phase, scales quadratically with the misalignment amplitude, and at a typical alignment of $\sigma = 10\ \mu$m exceeds the target sensitivity of $10^{-29}\,e$cm ($dS_y/dt \approx 1$ nrad/s) by three orders of magnitude. Using a symplectic spin-tracking simulation of the ring, we determine how far this systematic can be suppressed by orbit-based measurement and correction — the beam-based alignment (BBA) toolbox of modern accelerators. Standard orbit correction combined with simultaneous counter-rotating beams reduces the false EDM by a factor ~ 26, to about $6 \times 10^{-28}\,e$cm at $\sigma = 10\ \mu$m, with the residual scaling as σ^2. The residual is driven by the *symmetric* component of the misalignment pattern, in which neighboring quadrupoles kick the beam in opposite directions: its orbit signature is suppressed by roughly two orders of magnitude relative to the antisymmetric component. We show that this component cannot be recovered from the orbit at the required level by the *response-matrix and amplitude-readout* classes of techniques — K-modulation at a common or at per-quadrupole frequencies read as an amplitude, SVD inversion of the response matrix with or without regularization, neural-network estimators, an initial LOCO correction followed by drift monitoring, or harmonic fits — and that for these the limit is set not by beam-position-monitor (BPM) noise but by coherent systematics (optics "breathing" and β-beating) that do not average down, so that improved BPM technology, including SQUID-based BPMs, does not change the floor. The counter-rotating beam separation proposed as the control observable shares the same blindness. We further identify a *distinct* class — null-finding beam-based alignment, which locates each magnetic center by the zero-crossing of the local gradient-modulation response rather than by inverting a matrix or reading an amplitude — that is not subject to this wall and recovers the symmetric component in idealized optics; its robustness under realistic optics errors (β-beating) requires iterative orbit reduction and is established here for the vertical plane, with the horizontal plane and the hardware floor (magnetic-center motion under modulation) left as open items. Reaching $10^{-29}\,e$cm by the response-matrix classes therefore requires bounding the symmetric misalignment component a priori at the $\sim 1\ \mu$m level, which no orbit-based measurement can verify.

I. INTRODUCTION

A. The experiment: searching for a proton EDM

A permanent electric dipole moment (EDM) of a particle is a minute separation of its positive and negative charge distributions along the spin axis. A nonzero EDM requires the violation of time-reversal (hence CP) symmetry. Because the matter–antimatter asymmetry of the universe points to sources of CP violation much stronger than those contained in the Standard Model, finding a proton EDM above the Standard-Model expectation ($\lesssim 10^{-31}\,e$cm) would be direct evidence of new physics. The proposed storage-ring experiment [1] targets a sensitivity of $d = 10^{-29}\,e$cm.

The measurement principle is the *frozen-spin* method. Protons are stored at the "magic" momentum ($p \approx 0.7$ GeV/c) in a ring with electric bending and magnetic quadrupole focusing. At this momentum the spin remains locked to the momentum vector: it neither leads nor lags in the ring plane. If the proton carries an EDM, the radial electric field exerts a torque on it and the spin rotates *very slowly out of the ring plane*. The measured quantity is this rate,

$$f \equiv \frac{dS_y}{dt}, \qquad d = 10^{-29}\,e\cdot\text{cm} \;\leftrightarrow\; f \approx 1\ \text{nrad/s}, \tag{1}$$

one radian per billion seconds. The entire difficulty of the experiment follows from this number: every other effect capable of turning the spin out of plane at this level must either be eliminated or safely bounded. Effects that mimic the EDM are called *false EDM* signals, and this paper is about the most stubborn of them.

B. The false EDM: a geometric phase from misalignments

Each of the 48 quadrupole magnets of the ring will in practice be displaced from its ideal position by a few microns. The field at the center of a quadrupole vanishes; a beam passing off-center experiences a field proportional to the offset. The direction of the displacement determines what it does to the spin: a vertical offset dy exposes the beam to a radial field B_x, rotating the spin about the x axis; a horizontal offset dx exposes it to a vertical field B_y, rotating the spin about the y axis.

Either rotation alone is nearly harmless: averaged around the ring it largely cancels. The danger arises when both are present, because rotations about different axes *do not commute* — rotating first about x and then about y is not the same as the reverse — and the difference accumulates, turn after turn, as a small net rotation about the third axis: precisely the vertical precession channel in which the EDM is sought. This is an instance of the well-known geometric (Berry) phase. The consequence is

$$f_{\text{false}} \propto dx \cdot dy \implies f_{\text{false}} \propto \sigma^2, \tag{2}$$

i.e., the false EDM scales with the *square* of the rms misalignment σ (verified in Sec. II B: exponent $p = 2.00 \pm 0.01$, reproducing Fig. 9(a) of Ref. [1]). The numerical scale is unforgiving: a typical alignment of $\sigma = 10\ \mu$m produces a false EDM of $\sim 1000\times$ the target, living in the *same observable* dS_y/dt as the genuine signal.

The natural question is then: *is the beam not itself a measuring device?* The ring's 48 beam position monitors (BPMs) continuously read the closed orbit; misaligned quadrupoles distort that orbit; so measuring the misalignments from the orbit and correcting them — the standard *beam-based alignment* (BBA) practice of the accelerator community — should also clean up the false EDM. This paper quantifies how far this natural expectation holds, and where and why it stops.

C. What is already known

Most ingredients of the problem are, separately, well-solved problems; the contribution of this work is to combine them around the false-EDM question.

a. Orbit correction and observability. Global orbit correction using the singular-value decomposition (SVD) of the orbit response matrix — including the notion of poorly-observable ("decoupled") modes and their regularized rejection — goes back to Chung, Decker, and Evans [3]. The harmonic gain law used throughout this paper, $G_k = C/|Q^2 - k^2|$, is the harmonic limit of that framework. The systematics of biased estimators in degenerate orbit-response inversion were formalized by Wegscheider and Vilsmeier [4]; their degenerate parameter is the quadrupole strength, ours the transverse offset.

b. Beam-based alignment. Steering the beam through quadrupole magnetic centers is standard practice: varying a quadrupole's gradient and finding the beam position at which the orbit does not respond, accelerating the procedure with ac excitations [5], and performing it for many magnets simultaneously [6]. None of the measurement techniques examined in this paper is new in that sense; *what is new is the determination of their performance on the specific misalignment component that drives the false EDM.*

c. False-EDM budgets. The forward calculation misalignment $\to$ false EDM was performed for an all-electric ring in Ref. [2]; the full systematic budget of the magnetic-quadrupole symmetric-hybrid design is given by Omarov *et al.* [1], who show that the geometric-phase false EDM can be brought below target by the combination of simultaneous clockwise (CW) and counter-clockwise (CCW) beams, quadrupole polarity switching, and the reduction of the counter-rotating (CR) beam separation with dipole correctors.

d. Neighboring work. Circulant-response orbit correction in symmetric lattices [7]; on-line updating of the orbit response matrix [8]; the orbit imprint of coherent ground motion [9]; polarimeter performance and statistics [10].

D. The open gap

The control chain of Ref. [1] relies on *measuring* the CR beam separation and reducing it. However, (i) the measurement chain itself (48 BPMs or SQUID-based pickups plus K-modulation reconstruction) is not simulated there; (ii) a possible blindness to the misalignment component that is *invisible to the orbit* is not addressed; and (iii) whether polarity switching is a knob independent of the CW/CCW reversal is not quantified. These gaps are critical because — as shown below — the part of the misalignment that drives the post-correction false EDM is precisely the part least visible to the orbit.

E. This paper

Under realistic BPM systematics (offsets, gain errors, noise) and optics-model errors (β-beating, quadrupole tilt), we determine the false-EDM floor reachable by the orbit-based measurement-and-correction chain, as a function of the alignment tolerance: at $\sigma = 10\ \mu$m the floor is $\sim 6 \times 10^{-28}\ e$cm, scaling as σ^2. We show that the floor is *independent of BPM technology*: the limit is set not by white noise but by two coherent systematics that do not average down — optics "breathing" under per-quadrupole modulation, and β-beating combined with the conditioning of the response-matrix inversion — and we measure directly that the CR beam separation proposed as the control observable shares the same blindness.

Section II defines the simulation infrastructure, the false-EDM estimator, the symmetric/antisymmetric decomposition, and the systematic-error models. Section III first establishes the achievable suppression chain (the positive result), then presents the attempted methods for recovering the symmetric component from the orbit and where each stops. Section IV collects the conclusions with practical implications for the experiment.

II. METHOD

A. Simulation infrastructure: two layers

All numbers in this paper were produced with two complementary tools.

a. Full particle-and-spin tracker (C++). The ring — electric deflection sections and 24 FODO cells with 48 magnetic quadrupoles ($R_0 = 95.49$ m, $g \approx 0.2$ T/m, vertical tune $Q_y \approx 2.3$) — is integrated element by element through the actual fields with a fourth-order Gauss–Legendre *symplectic* integrator; the spin is propagated in the same step by the Thomas-BMT equation. Symplecticity guarantees that energy and amplitude errors do not accumulate over long tracking, which is what makes nrad/s-level slopes measurable at all. The genuine EDM enters through a switch ($\eta = 1.88 \times 10^{-15} \to dS_y/dt = 9.81 \times 10^{-10}$ rad/s, odd under the CW/CCW difference, in exact agreement with Eq. (C1) of Ref. [1]); all false-EDM measurements are taken with the EDM switch *off*, so that any measured vertical precession is by definition systematic.

b. Analytic Twiss / closed-orbit solver. From the 2×2 transfer matrices of the quadrupoles and drifts, the lattice functions $\beta(s)$, $\phi(s)$, the tune Q, and the orbit response matrix

$$R_{ij} = \frac{\sqrt{\beta_i \beta_j}}{2 \sin \pi Q} (KL)_j \cos\bigl(|\phi_i - \phi_j| - \pi Q\bigr) \qquad (3)$$

are built in closed form. This layer runs in seconds and is used for wide scans (noise Monte Carlo, β-beating sweeps); it agrees with the C++ tracker to within 1%, and the two codes share nothing but the configuration file, so their agreement cannot come from a common modeling error.

c. A methodological caution. In inverse-problem analyses the response matrix must be built from *real beam dynamics*, not from idealized Twiss formulas: the idealized R can hide the smallest-singular-value (unobservable) mode. In this work the analytic R shows a condition number of 193 while the beam-dynamics R contains an additional near-null mode at the 10^{-7} level.

B. Measuring the false EDM correctly

The goal looks simple: the slope of $S_y(t)$. The difficulty is that the betatron oscillation around the closed orbit feeds oscillatory components into the spin that exceed the sought secular slope by orders of magnitude; a naive linear fit mistakes them for signal. Two measures are used together:

1. *4D closed orbit.* The tracked particle is placed, by Newton iteration in (x, x', y, y'), at the point of vanishing betatron variance — exactly on the closed orbit — so there is no betatron amplitude to cancel in the first place.
2. *Model fit.* From $S_y(t) = a + bt + \sum_k [c_k \cos \omega_k t + d_k \sin \omega_k t]$ only the secular slope b is retained; residual oscillatory content is absorbed by the fit. (A plain polynomial fit is *never* used.)

The estimator was validated by three independent tests:

- **σ^2 test:** scanning $\sigma = 2.5/5/10$ μm gives $|f| \propto \sigma^p$ with $p = 2.00 \pm 0.01$ [Fig. 1] — pure quadratic geometric phase, no linear leakage, reproducing Fig. 9(a) of Ref. [1].
- **Sign test:** under sign flips of the misalignment pattern, $f(+dx, -dy) = -f(+dx, +dy)$ and $f(-dx, -dy) = +f(+dx, +dy)$ hold to the 10^{-3} level — the measured quantity is purely *bilinear* $(dx \cdot dy)$.
- **Four-particle equivalence:** the averaged $S_y(t)$ of four symmetric starts placed *around the closed orbit*, $v_{\rm CO} \pm (\pm\delta, \pm\delta)$ with $\delta = 50$–200 μm, yields the same slope as the single on-orbit particle to within 1%.

a. A trap worth reporting. If the four symmetric starts are placed around the *ideal axis* (i.e., without knowledge of the closed orbit), the method fails: the $\pm\delta$ betatron oscillations cancel pairwise in the average, but the betatron component induced by the closed-orbit offset itself is *common* to all four particles and survives. The measurement then collapses to the same betatron artifact as a single off-orbit particle (-3.0×10^{-4} rad/s, $\sim 300\times$ the signal). Symmetric sampling does not substitute for closed-orbit knowledge; it works around it.

C. Two faces of a misalignment pattern: symmetric and antisymmetric

All results of this paper rest on one decomposition, defined here explicitly. Write the 48 quadrupole offsets as a vector v, with QF at index $2k$ and QD at $2k{+}1$ in cell k. Separate, in each cell, the amounts by which the two quadrupoles move *together* and *oppositely*:

$$\begin{aligned}(P_{\rm sym}\, v)[2k] &= (P_{\rm sym}\, v)[2k{+}1] = \tfrac{1}{2}\bigl(v[2k] + v[2k{+}1]\bigr),\\ (P_{\rm anti}\, v)[2k] &= -(P_{\rm anti}\, v)[2k{+}1] = \tfrac{1}{2}\bigl(v[2k] - v[2k{+}1]\bigr),\end{aligned} \qquad (4)$$

with $P_{\rm sym} + P_{\rm anti} = I$ and $P_{\rm sym} P_{\rm anti} = 0$.

The physics of the split is this: QF and QD have gradients of *opposite sign.* Two neighboring quadrupoles displaced in opposite directions (the antisymmetric part) kick the beam in the *same* direction — the kicks add up into a smooth, low-azimuthal-harmonic pattern close to the betatron resonance, and a *large* closed-orbit distortion results. Two neighbors displaced in the same direction (the symmetric part) kick *oppositely* — the kicks

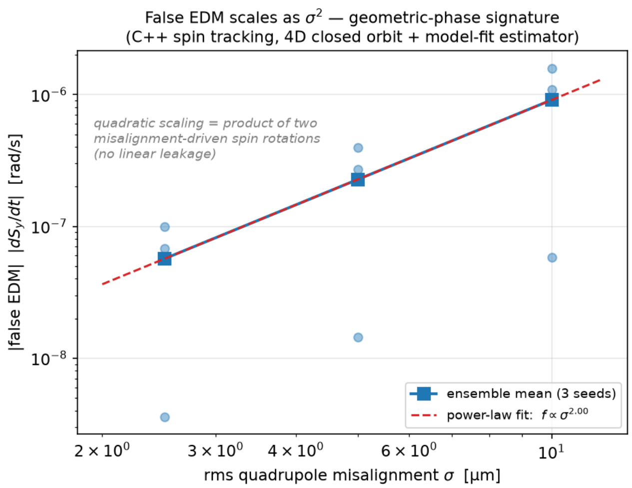

FIG. 1. Validation of the false-EDM estimator: the measured $|dS_y/dt|$ scales as $\sigma^{2.00}$ across $\sigma = 2.5$–10 μm (C++ tracking, three seeds per point), the signature of a pure geometric phase with no linear leakage. The per-seed scatter reflects the bilinear character of f: individual patterns can conspire to small values.

cancel within the cell, and what remains is a cell-to-cell alternating pattern at $k \approx 24$, crushed by the orbit gain

$$G_k = \frac{C}{|Q^2 - k^2|}, \qquad C \approx 24.8,\ Q^2 \approx 5 \tag{5}$$

since $Q \approx 2.3 \ll 24$ [Fig. 2, right].

This is not an assumption but a property of the response matrix itself, demonstrated directly [Fig. 2, left]: computing the symmetric content $\|P_{\rm sym} v_i\|^2$ of each right-singular vector of $R = U\Sigma V^\top$ shows the *large*-singular-value modes to be nearly purely antisymmetric and the *small*-singular-value modes nearly purely symmetric; the visibility gap between the two families is summarized by cond$(R) = 193$. Concretely, the same 10-μm-rms pattern leaves a $\sim$115 μm orbit trace if antisymmetric but only $\sim$1.7 μm if symmetric.

The whole of Sec. III is built on the resulting tension: *the part of the misalignment that drives the post-correction false EDM is the symmetric one — the part least visible to the orbit.*

D. How success is measured: the correlation trap

Two metrics are used for reconstruction quality; their difference will be decisive several times, so we define both explicitly.

a. Pearson correlation. Between reconstruction $\hat v$ and truth v, over 48 components, corr $= \sum_j (\hat v_j - \langle \hat v\rangle)(v_j - \langle v\rangle)/(48\,\sigma_{\hat v}\sigma_v)$. This common metric is misleading here for two reasons: (i) it is scale- and offset-invariant — the mean subtraction in its definition removes precisely the worst (uniform) member of the symmetric family from the error; (ii) it is dominated by the many well-reconstructed antisymmetric components. As a result corr $= 0.99$ *is compatible with a completely wrong symmetric component* (a concrete example appears in Fig. 4).

b. Component-error metric (the correct one). The rms, *without* mean subtraction, of the symmetric and antisymmetric parts of the error vector $e = \hat v - v$: $\|P_{\rm sym}\, e\|_{\rm rms}$ and $\|P_{\rm anti}\, e\|_{\rm rms}$, compared against the corresponding components of the signal itself (both ≈ 41 μm for ± 100 μm uniform misalignments). A method "sees" the symmetric component only if its symmetric error is below the symmetric signal.

E. Systematic error models

Four systematics enter the model at realistic magnitudes:

- **BPM offsets** (static, ~ 100 μm per BPM; variant 30 μm): each BPM's electronic zero differs from the true axis by an unknown constant, stable over hours. The prime killer of single-orbit inversion; cancels in difference-based measurements (K-modulation, drift).
- **BPM noise** (white, $\sim$1 μm single shot): averages down as $\sqrt{N}$; with synchronous detection it ceases to be a practical limit. One of the main messages of this paper is that the real limit is *not* this.
- **BPM gain errors** (multiplicative, $\sigma_g = 1$–10% scans): percent-scale errors of the readings; partially softened by multi-BPM averaging.
- **β-beating / gradient model error** (per-quadrupole $\varepsilon = 0.5$–5%, applied in the tracker as per-quadrupole gradient deviations): represents the difference between the real machine optics and the model used in reconstruction (typical LOCO-calibrated residual $\sim 1\%$). A *coherent* model systematic that does not average down; it will prove to be the decisive wall of the inverse methods.
- **Quadrupole tilt** (roll angle ψ): two distinct channels, cross-plane leakage into the measurement and a direct geometric-phase contribution to the false EDM; the latter dominates and is reported as a separate tolerance item.

a. Seed policy. The bilinear f can be accidentally small or large for a single misalignment realization; every claim is therefore reported over multi-seed ensembles (typically 3–40 realizations) and no root cause is inferred from a single-seed result.

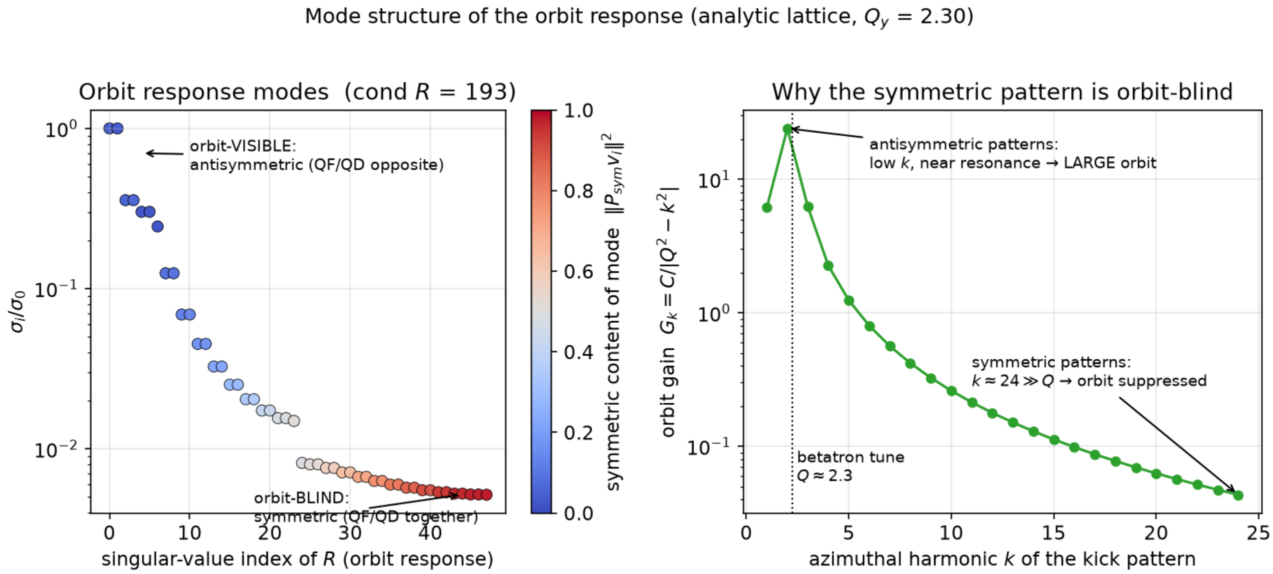

FIG. 2. Mode structure of the orbit response (analytic lattice, $Q_y = 2.30$). Left: singular values of R, colored by the symmetric content of the corresponding misalignment mode; large-singular-value (orbit-visible) modes are antisymmetric, small-singular-value (orbit-blind) modes are symmetric; cond$(R) = 193$. Right: the orbit gain law $G_k = C/|Q^2 - k^2|$; antisymmetric patterns produce low-k kicks near the betatron resonance, symmetric patterns produce $k \approx 24$ kicks far above it.

III. SIMULATION RESULTS

A. The achievable suppression chain

We first establish what orbit-based control *does* deliver. For white misalignments of $\sigma = 10$ μm in both planes the raw false EDM is $|f| \approx 10^{-6}$ rad/s, about $1000\times$ target [Eq. (1)]. Two standard measures reduce it (Table I, Fig. 3):

1. *Simultaneous counter-rotating beams.* The genuine EDM is odd under beam reversal [Eq. (C1) of [1]], so the CW/CCW difference cancels the part of the false EDM that is even. Measured gain: a factor 3.4 (ensemble median), leaving $474\times$ target.

2. *Orbit correction.* Flattening the closed orbit with correctors removes the orbit-visible (antisymmetric) part of the misalignment imprint. Measured additional gain: $7.7\times$, leaving **62× target** (6.05×10^{-8} rad/s $\approx 6 \times 10^{-28}$ $e$cm).

The residual after both steps lives almost entirely in the symmetric, orbit-blind subspace of Sec. II C, and by Eq. (2) it scales as σ^2: the central deliverable of this paper is therefore the curve of Fig. 3. Reaching the target along this curve requires the *symmetric* misalignment component to be at or below $\sigma_{\rm sym} \approx 1.3$ μm. Two facts make that requirement problematic. First, as the rest of this section shows, no orbit-based measurement can verify it. Second, mechanical surveying cannot verify it either, because the relevant quantity is the *magnetic* center: stray dipole components at the $\sim$20 μT level displace the magnetic center from the mechanical one by tens of microns, and only the beam sees the magnetic center.

TABLE I. The suppression chain at $\sigma = 10$ μm (C++ tracking). The equivalent EDM uses 1 nrad/s $\leftrightarrow 10^{-29}$ $e$cm.

Stage	$\|f\|$ / target	equiv. EDM [$e$cm]
raw (no correction)	$\sim 1000\times$	$\sim 10^{-26}$
+ CW/CCW difference	$474\times$	$\sim 5 \times 10^{-27}$
+ orbit correction	$62\times$	$\sim 6 \times 10^{-28}$
+ any method of Sec. III B	$62\times$	floor

B. Attempts to measure the symmetric component from the orbit

Everything in this subsection belongs to one class: reconstruct the misalignments (or their symmetric component) from orbit readings, then correct. Five independent implementations hit the same wall.

1. Common-frequency K-modulation with matrix inversion

Modulating all quadrupole gradients together between two settings (g and $1.02g$) and taking the difference of the two closed orbits removes the static BPM offsets exactly and yields $\Delta y = \Delta R\, \Delta q$ with $\Delta R = R(1.02g) - R(g)$. Inverting the full 48×48 matrix reconstructs the offsets perfectly in the absence of errors (corr = 1.000000);

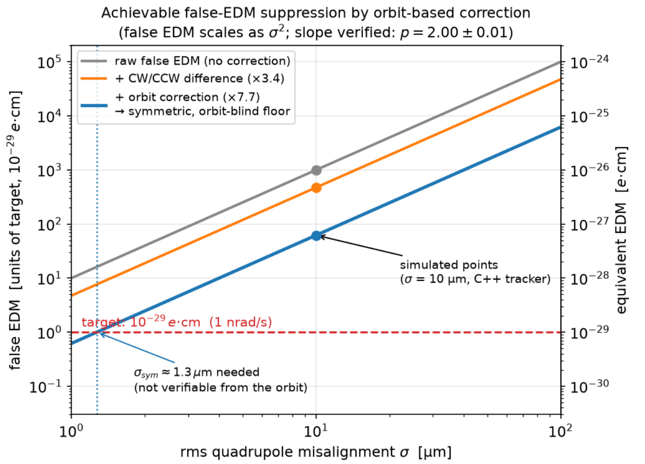

FIG. 3. Main result: the false EDM reachable by orbit-based correction as a function of the rms quadrupole misalignment. Points at $\sigma = 10$ μm are C++ tracking results; all curves scale as σ^2 (verified exponent $p = 2.00$). The corrected floor crosses the target only at $\sigma_{\rm sym} \approx 1.3$ μm — a level that no orbit-based measurement can verify.

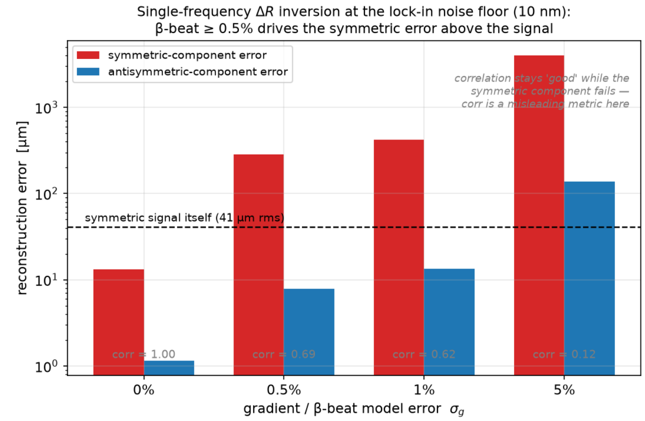

FIG. 4. Single-frequency ΔR inversion at the lock-in noise floor (10 nm, 40 seeds). Symmetric- and antisymmetric-component reconstruction errors vs. the optics-model error σ_g; the dashed line is the symmetric signal itself. Any realistic model error ($\geq 0.5\%$) drives the symmetric error far above the signal while the Pearson correlation (gray labels) remains deceptively high — the correlation trap of Sec. II D.

optics breathing (below) is no obstacle here because it is contained in ΔR itself. The obstacle is conditioning: $\mathrm{cond}(\Delta R) = 3.7 \times 10^4$ with $\sigma_{\rm min} \approx 10^{-4}$, and the small-singular-value directions are the symmetric ones.

White BPM noise *can* be beaten: with the modulation at $\sim$1 kHz and synchronous (lock-in) detection the effective noise falls as $\sqrt{N}$, reaching $\sim$14 nm in $\sim$17 minutes. The genuine limits are the ones that do not average [Fig. 4]: even at a 10 nm effective noise floor with a *perfect* optics model the symmetric-component error is already a third of the symmetric signal; with a realistic optics-model error the inversion collapses — $\varepsilon = 0.5\%$ of β-beating drives the symmetric error to $\sim 7\times$ the signal (283 μm vs. 41 μm) while the correlation still reads a deceptively healthy 0.69. Because β-beating is a coherent systematic, no BPM technology — including SQUID-based BPMs, whose low noise would only shorten the integration time — changes this floor.

a. An offset-cancellation lower bound. The pattern above is not an accident of this scheme. Cancelling the static BPM offsets by any two-setting difference means that the available information is no longer R but $\Delta R \sim \varepsilon_g\, \partial R/\partial g$ for a relative setting difference ε_g; its inverse grows as $\|\Delta R^{-1}\| \sim \|R^{-1}\|/\varepsilon_g$. Within the natural class of linear, unbiased estimators that cancel static offsets exactly by a two-setting difference, offset immunity is thus purchased at the price of a $1/\varepsilon_g$ noise amplification. We state this as a structural lower bound for that class — estimators outside it (e.g., partial Bayesian modeling of the offsets) are not covered by the argument.

2. *Per-quadrupole K-modulation at distributed frequencies: optics breathing*

A seemingly stronger idea avoids matrix inversion altogether. Modulate each quadrupole at its *own* frequency ($\sim$2% depth); the amplitude A_i of frequency f_i in a BPM signal should be proportional to the beam–quadrupole offset o_i, giving a direct per-quadrupole readout — the ac-BBA concept of Ref. [5] transplanted to this ring. Since the modulation (1–10 kHz) is far below the betatron frequency ($\sim$ 515 kHz), the beam adiabatically follows the instantaneous closed orbit and each amplitude equals a static derivative $\partial y_{\rm BPM}/\partial g_i$, computable from two static closed orbits.

The method fails for a reason worth understanding [Fig. 5]. Modulating quadrupole i does two things: it modulates the quadrupole's own feed-down kick (the sought signal, $\sim$0.9 μm at the BPM), and it modulates the *one-turn matrix* — hence $\beta(s)$, $\phi(s)$, and Q everywhere. The pre-existing closed orbit ($\sim$ 0.37 mm rms, built by all 48 offsets together) is re-transported through this modulated optics, producing a few-μm response at the BPM: the optics "breathes." The breathing-to-signal ratio is ~ 7 ($S/B \approx 0.14$). A decisive decomposition: the response to modulating one quadrupole is -9.83 μm with the full misalignment pattern present, but only -0.037 μm when only that quadrupole's own offset is kept — a factor 266. The demodulated amplitude measures *the orbit built by the neighbors*, not the quadrupole's own offset; dividing by a calibration is meaningless.

Because the breathing term is *coherent* (approximately the existing orbit times a scalar), it does not average away: going from 1 to 48 BPMs leaves the reconstruction correlation at zero ($+0.07 \to -0.03$), and a SQUID's low noise is of no help, since noise was never the problem. In a breathing-free idealization the same pipeline reconstructs perfectly (corr = 1.000), pinning the failure unambiguously on the breathing. The C++ tracker confirms the analytic breathing amplitudes to $\sim 1\%$. BPM offsets, incidentally, are irrelevant in this branch — ac demodulation suppresses the static offset exactly when the modulation frequencies are locked to the integration window — so improving offsets from 100 to 30 μm changes nothing.

We emphasize the conceptual distinction between the two K-modulation failures: breathing is a coherent *contamination* that kills the per-quadrupole readout; conditioning is a *noise-amplification* property of the inversion. They are different walls, and each closes its branch.

3. Neural-network reconstruction

Could a learned estimator outperform the linear algebra? No — and the reason is informative. The map misalignment → orbit is *exactly linear* ($y = R\,\Delta q$), so a network trained on it can only relearn R; the inverse direction is then R^{-1} with the same conditioning. Numerically, a multilayer perceptron (two hidden layers of 128, 4000 training samples, 1 μm BPM noise) reconstructs the antisymmetric component as well as truncated SVD (0.71 vs. 0.63 μm error) and fails on the symmetric component exactly as badly (5.6 vs. 6.3 μm error against a 9.9 μm signal). The limit is information-theoretic: the symmetric imprint is below the noise-plus-systematics floor, and no estimator can extract information that is not in the data. For the same reason more expressive architectures (convolutional, attention-based, graph networks) cannot help — capacity is not the bottleneck in a linear problem; and for white noise the matched filter (lock-in) is already optimal, so learned time-series estimators (LSTMs) cannot beat it either. Any real gain would have to come from *prior information*; sparsity priors (LASSO, CLEAN, greedy harmonic selection) were tried and hit the same observability floor.

4. LOCO-style calibration plus drift monitoring

A pragmatic strategy accepts that absolute alignment cannot be measured and asks only for its *changes*: calibrate the machine once (LOCO), then monitor the orbit difference $y(t) - y_0$, in which the static BPM offsets cancel exactly. This works well — for what it sees: drifts are tracked with 6.5 μm rms accuracy under 50 μm BPM offsets (a 29× improvement over the absolute reconstruction), degrading only mildly with 1% β-beating. But the difference orbit passes through the same response matrix: the eight worst-observable modes are 96% symmetric with a 193× noise amplification. The drift monitor is a useful, cheap watchdog for the antisymmetric part — and is blind to the symmetric part like everything else in this class.

5. Harmonic approaches

Two harmonic-domain variants deserve a note. *Targeted Fourier reconstruction:* if the harmonic content of the misalignment is known in advance, restricting the basis improves the conditioning from $\sim$13000 to 186 and reconstruction works; but for unknown patterns greedy or LASSO harmonic selection collapses from rank deficiency — knowing *which* harmonics is precisely the unavailable information. *Harmonic fit of a known signature:* reading a known field imprint (e.g., an N=2 stray radial field, which drives a vertical k=2 orbit) by projecting the 48-BPM data on $\cos 2\theta / \sin 2\theta$ is perfectly conditioned but *biased*: the k=2 coefficient sums the field, the k=2 content of the misalignments, and the k=2 content of the BPM offsets, which one number cannot separate — a $\sim$53 nT bias at 10 μm misalignments against a 1 nT target in our case study. Full SVD-with-regularization does not beat it under realistic offsets: after gain calibration of the regularized estimator its detection floor is $\sim$45 nT at 100 μm offsets (2.6 nT only for ideal, offset-free BPMs). In the language of Ref. [4]: the forward projection is bias-dominated, the inversion is variance-dominated, and neither reaches the degenerate information.

6. The counter-rotating beam separation

Reference [1] proposes controlling the geometric phase by measuring and zeroing the *separation* of the CW and CCW closed orbits. The separation, however, is itself an orbit difference, and inherits the orbit's mode structure. We measured this directly [Fig. 6]: for 10-μm-rms vertical patterns the ratio of the antisymmetric to the symmetric signature is 5.7× in the single-beam closed orbit and 8.0× in the CR separation (three seeds; an independent pattern set gave 3.8× and 4.5×). The suppression factors are similar: *the CR separation opens no additional window on the symmetric component.* Measuring and zeroing it controls the orbit-visible part of the geometric phase — consistent with Ref. [1] — and leaves the symmetric residual of Table I untouched. This closes gap (ii) of the Introduction.

C. Attempts to open the symmetric window

Two lattice-level escapes were examined and quantified.

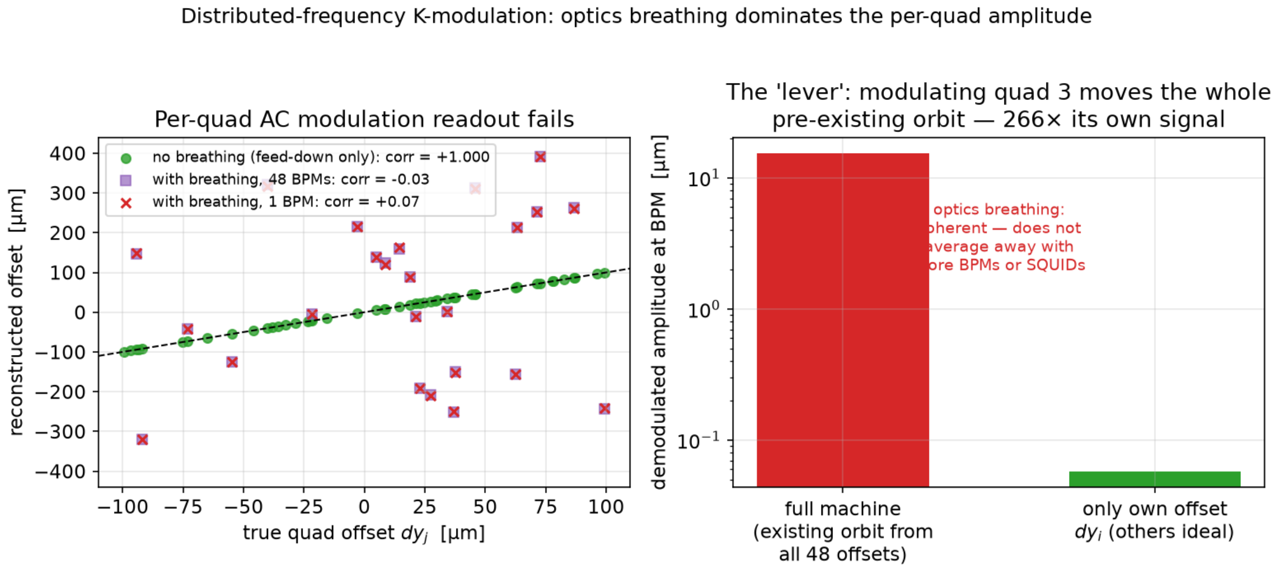

FIG. 5. Distributed-frequency per-quadrupole K-modulation. Left: reconstructed vs. true offsets — perfect without breathing (green), zero correlation with breathing included, for 1 BPM (red) and even for 48 BPMs (purple). Right: the "lever" — the demodulated amplitude responds to the whole pre-existing orbit (266× the quadrupole's own contribution); being coherent, this contamination does not average away with more BPMs or better (SQUID) resolution.

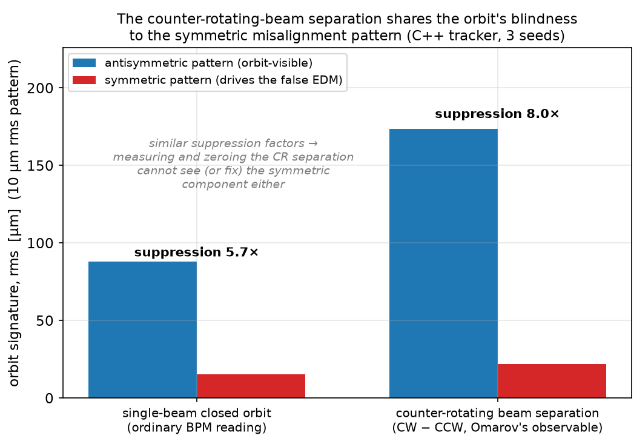

FIG. 6. The counter-rotating beam separation (right group) suppresses the symmetric pattern by a factor similar to the ordinary single-beam orbit (left group): Omarov's control observable shares the orbit's blindness to the component that drives the residual false EDM.

1. *Quadrupole polarity switching*

Reversing all quadrupole currents ($g \to -g$) reverses each misalignment-driven spin rotation, so one might hope it reverses the false EDM while leaving the genuine (electric-field-driven) EDM intact. It does not: the geometric phase is the *product of two* field-driven rotations, each $\propto g$, so f is *even* in g. Measured flip ratios $f(-g)/f(+g)$ are +2.8, +0.85, and +1.17 for symmetric, antisymmetric, and generic patterns respectively — the sign never reverses. Moreover, in the idealized periodic lattice the polarity flip is *degenerate* with the beam-direction reversal: $f(\mathrm{CCW}, +g) = f(\mathrm{CW}, -g)$ to a measured ratio of 1.00, so the four-fold combination of Eq. (C2) of Ref. [1] collapses to the two-fold CW/CCW difference, and polarity switching provides no independent cancellation knob for the geometric phase. (It remains useful against systematics that are odd in g.) This closes gap (iii) of the Introduction.

2. *Raising the betatron tune*

The symmetric pattern is hidden because its kick harmonic ($k \approx 24$) lies far above the tune [Eq. (5)]; raising Q toward k should make it visible. This partially works: at $Q_y \approx 10.3$ (alternating focusing preserved, machine stable) the high-m part of the symmetric family reaches orbit gains comparable to the antisymmetric ones. But the low-m part requires $Q \geq 16$, while the 180°-per-cell stopband caps the tune at $Q \leq 12$; only ∼8–25% of the symmetric content becomes recoverable, worth only a factor ∼2 in false-EDM suppression.

More decisively, the price of a high tune is *strong quadrupoles*: Q_y : 2.3 → 10.3 requires g : 0.21 → 0.69 T/m. Stronger gradients kick the beam harder at the same misalignment; since the geometric phase is a product of two such rotations ($\propto g^2$), with an additional contribution from the enlarged betatron orbit, the false EDM grows steeply — a single-pattern C++ measurement gave a factor ~ 32 ($\sim g^3$) from $g = 0.21$ to 0.69 T/m — while the genuine EDM signal, generated by the electric deflectors, is independent of g. The signal-to-background thus *worsens* by the same large factor: one cannot operate at high tune, and a temporary high-tune diagnostic transfers at most the recoverable $\sim$25% back to the operating point. (We note that at $Q \approx 10$ the closed-orbit determination itself becomes delicate; our ensemble scans at $g = 0.69$ T/m carried residual-betatron contamination and we therefore quote only the robust qualitative conclusion and the single clean pattern above.)

D. Outside the orbit: off-momentum spin sensing (brief)

For completeness we record one attempted escape outside the orbit toolbox. Storing the beam slightly off the magic momentum makes the in-plane spin precess at $\nu_s \neq 0$, and the geometric-phase contribution to the vertical spin oscillation is indeed *amplified* $\propto 1/\nu_s$ — suggesting a fast, resonant readout of the false EDM. The idea fails because misalignments also tilt the invariant spin axis at *first* order in σ: this $\mathcal{O}(\sigma)$ oscillation sits at the same frequency ν_s and exceeds the $\mathcal{O}(\sigma^2)$ geometric-phase term by a factor ~ 3600 at $\sigma = 10$ μm. Every separation channel was tested and closed: the two terms do not separate in $\delta p/p$, their relative phase is pattern-dependent (91° spread), four-fold sampling does not cancel the first-order term (it is common to the four particles), the amplitude crossing lies at an unphysical $\sigma \approx 62$ mm, a full 3D fit of the invariant axis leaves no residual to fit, and the longitudinal component is knob-insensitive off-magic. The magic momentum is the *unique* working point in that it converts the dominant first-order effect into a harmless constant offset — which is also why the estimator of Sec. II B works there.

IV. CONCLUSIONS

a. Quantitative result. The orbit-based chain — simultaneous counter-rotating beams, standard orbit correction, and continuous drift monitoring — bounds the quadrupole-misalignment false EDM at $\sim 6 \times 10^{-28}$ $e$cm for $\sigma = 10$ μm rms misalignments, and the bound scales as σ^2 (Fig. 3). If the required suppression were only to the 10^{-27} $e$cm level, existing BBA technology would thus suffice. Reaching 10^{-29} $e$cm requires the *symmetric* component of the magnetic-center misalignments to be at the ~ 1.3 μm level.

b. Wall for the response-matrix and amplitude classes. The symmetric component is unobservable at the required level in every channel of these classes that we tested: common-frequency K-modulation with (regularized) matrix inversion, distributed-frequency per-quadrupole modulation read as an amplitude, neural-network estimators, LOCO-plus-drift monitoring, harmonic fits, and the counter-rotating beam separation. For these the limit is set not by white BPM noise — which synchronous detection beats — but by two coherent systematics that do not average down: optics breathing under per-quadrupole modulation, and β-beating amplified by the conditioning of the inversion; and by the lattice itself, whose stopband caps the tune far below the symmetric kick harmonic. *For these classes no BPM technology, including SQUID-based BPMs, changes the floor.* Polarity switching provides no independent cancellation (it is degenerate with beam reversal, and the geometric phase is even in g), and mechanical metrology cannot substitute because the driving quantity is the beam-defined magnetic center.

c. A distinct class that overcomes the wall: null-finding BBA. The walls above are specific to inverting a response matrix or dividing a demodulated amplitude by a calibration. Standard quadrupole beam-based alignment (BBA) does neither: it steers the beam through each quadrupole and locates the magnetic center as the beam position at which the gradient-modulation response *vanishes*. Because a zero-crossing is independent of the response slope, this measurement is not amplified by the conditioning and is, in idealized optics, transparent to β-beating; it carries no "symmetric blindness." In our tracking it measures the symmetric centers and suppresses the false EDM from $\sim 356\times$ to $\sim 1.6\times$ target in idealized optics. Under realistic 1% β-beating the naive single-pass measurement fails, because modulating a quadrupole re-transports the large uncorrected orbit (optics breathing) and β-beating makes this bias mode-dependent; correcting the orbit first and iterating restores it in the vertical plane (residual 0.14 μm, the clean-optics floor). The horizontal plane, where a few quadrupoles near the lattice seam are corrupted by dispersion, and the hardware floor set by any magnetic-center motion under gradient modulation, remain open. We therefore present null-finding BBA not as a completed solution but as the one measurement class that is *not* excluded by the walls above — an open and promising route to the symmetric component.

d. Scope. Active spin-based nulling is outside the scope of this paper for a fundamental reason: the false and genuine EDM live in the same observable dS_y/dt, so a feedback that nulls the measured precession nulls the signal as well; the only discriminator is the CW/CCW difference, which is precisely what leaves the $62\times$ residual above. (Its statistics are also prohibitive: polarimetry reaches the nrad/s level only on year timescales [10].) Forward-predicting the false EDM from the measured orbit — as opposed to reconstructing misalignments — is a separate open problem not covered by the present limit.

e. Implications for the experiment. Our results support the control philosophy of Ref. [1] while sharpening its boundaries: (a) the CW/CCW combination and orbit-based separation control handle the orbit-visible part of the geometric phase, and a standard-BPM orbit monitor is an inexpensive substitute for SQUID-based pickups for that part (a $7.7\times$ gain), with drift monitoring providing a free, continuous watchdog; (b) the symmetric residual must be controlled *a priori* — by construction tolerances on the magnetic-center alignment and by field quality — because it can be neither measured nor actively nulled afterwards; (c) the σ^2 law makes this tolerance requirement concrete: every factor of two in symmetric alignment buys a factor of four in false EDM.

f. Validity. All numbers refer to one lattice (24 FODO cells, $Q_y \approx 2.3$) in a linear model (no sextupoles); the suppression factors (3.4×, 7.7×, 62×) carry finite ensemble statistics; and the quadrupole roll angle constitutes a separate direct channel with a ∼ 0.3 mrad tolerance, outside the misalignment budget quantified here. The qualitative structure — the symmetric/antisymmetric visibility gap and the coherent-systematic walls — follows from the alternating- gradient principle itself and is expected to carry over to any strong- focusing implementation.

ACKNOWLEDGMENTS

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